1. Create an HTML document with the following formatting options:
(a)Bold,
(b) Italics,
(c) Underline,
(d) Headings (Using H1 to H6 heading styles),
(e)Font (Type, Size and Color),
(f) Background (Colored background/Image in background),
(g) Paragraph,
(h) Line Break,
(i) Horizontal Rule,
(j) Pre tag

```
<html>

 <head>

<title>HTML Formatting Example</title>

</head> <body bgcolor=”blue”>

<!-- Headings -->

 <h1>Heading 1</h1>

<h2>Heading 2</h2>

 <h3>Heading 3</h3>

<h4>Heading 4</h4>

<h5>Heading 5</h5>

<h6>Heading 6</h6>

<!-- Paragraph -->

<p>This is a paragraph demonstrating text formatting in HTML.</p>

 <!-- Bold -->

<p><b>This text is bold.</b></p>

<!-- Italics -->

 <p><i>This text is italic.</i></p>

<!-- Underline -->
```

```
<p><u>This text is underlined.</u></p>

<!-- Font (Type, Size, Color) -->

 <font face="monotype corsiva" size=13 color="blue">This text uses a custom font, size, and color.</font>

<!-- Line Break -->

<p>This is the first line.<br>This is the second line after a line break.</p>

<!-- Horizontal Rule -->

<hr>

 <!-- Preformatted Text -->

<pre> This is preformatted text.

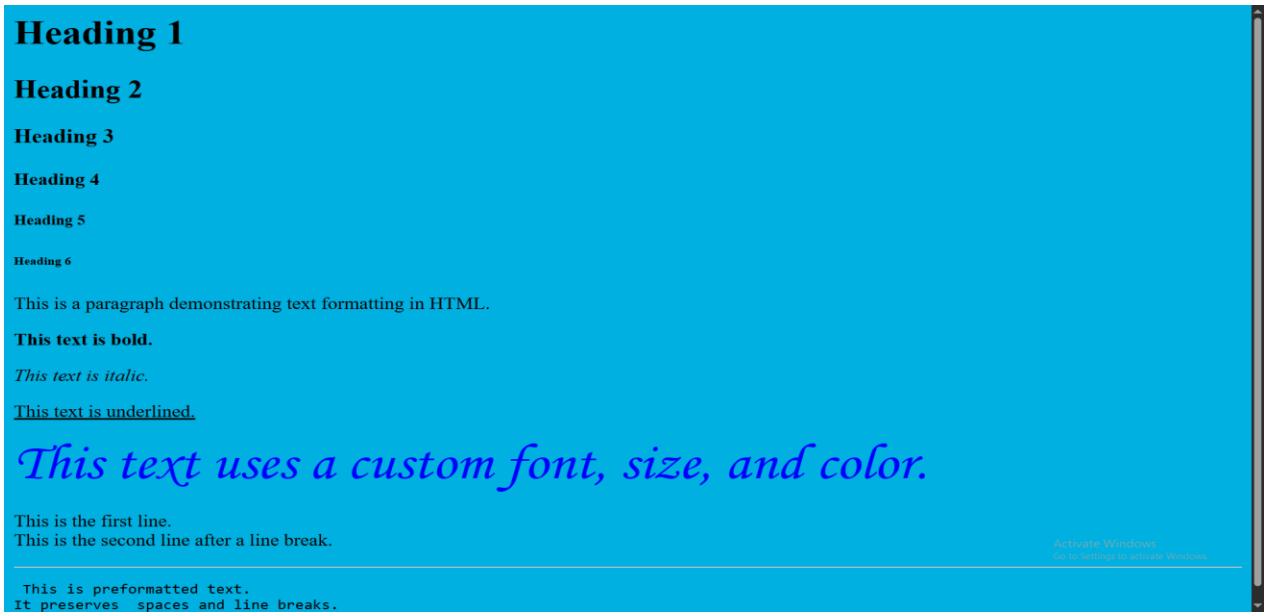
It preserves  spaces and line breaks.

</pre>

 </body>

</html>
```

OUTPUT:



2. 2. Create an HTML document which consists of:
(a) Ordered List
(b) Unordered List
(c) Nested List
(d) Image

```
<html>
<head>
<title>HTML Lists and Image Example</title>
</head>
<body>
<h1>HTML List and Image Example</h1>
<!-- Ordered List -->
<h2>Ordered List</h2>
<ol>
<li>B.Com</li>
<li>B.Sc.,</li>
<li>BCA</li>
<li>BBA</li>
</ol>
<!-- Unordered List -->
<h2>Unordered List</h2>
<ul>
<li>B.Com</li>
<li>B.Sc.,</li>
<li>BCA</li>
<li>BBA</li>
```

```
</ul>
<!-- Nested List -->
<h2>Nested List</h2>
<ul>
<li>B.Sc.,
<ul>
<li>Science Wing</li>
<li>Life Sceince Wing</li>
</ul>
</li>
<li>B.Sc.,
<ol>
<li>Mathematics</li>
<li>Pyhsics</li>
<li>Electronics</li>
<li>Statistics</li>
<li>Computer Science</li>
</ol>
</li>
</ul>
<!-- Image -->
<h2>Image Example</h2>
<img src="URL" alt="Sample Image">
</body>
</html>
```

OUTPUT:

HTML List and Image Example

Ordered List

1. B.Com
2. B.Sc.,
3. BCA
4. BBA

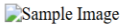
Unordered List

- B.Com
- B.Sc.,
- BCA
- BBA

Nested List

- B.Sc.,
  - Science Wing
  - Life Sceince Wing
- B.Sc.,
  1. Mathematics
  2. Pyhsics
  3. Electronics
  4. Statistics
  5. Computer Science

Image Example



Activate Windows
Go to Settings to activate Windows.

3. Create a Table with four rows and five columns. Place an image in one column.

```
<html>
<head>
<title>HTML Table with Image</title>
<style>
table
{
width: 80%;
border-collapse: collapse;
margin: 20px 0;
}
th, td
{
border: 1px solid #333;
padding: 10px;
text-align: center;
}
img
{
max-width: 100px;
height: auto;
} </style>
</head>
<body>
```

```
<h1>Table with Image</h1>
<table>
<tr>
<th>Header 1</th>
<th>Header 2</th>
<th>Header 3</th>
<th>Header 4</th>
<th>Header 5</th>
</tr>
<tr>
<td>Row 1, Col 1</td>
<td>Row 1, Col 2</td>
<td>Row 1, Col 3</td>
<td>Row 1, Col 4</td>
<td><img src="URL" alt="Sample Image"></td>
</tr>
<tr>
<td>Row 2, Col 1</td>
<td>Row 2, Col 2</td>
<td>Row 2, Col 3</td>
<td>Row 2, Col 4</td>
<td>Row 2, Col 5</td>
</tr>
<tr>
<td>Row 3, Col 1</td>
```

```
<td>Row 3, Col 2</td>
<td>Row 3, Col 3</td>
<td>Row 3, Col 4</td>
<td>Row 3, Col 5</td>
</tr>
</table>
</body>
</html>
```

OUTPUT:

Table with Image

Header 1	Header 2	Header 3	Header 4	Header 5
Row 1, Col 1	Row 1, Col 2	Row 1, Col 3	Row 1, Col 4	Sample Image
Row 2, Col 1	Row 2, Col 2	Row 2, Col 3	Row 2, Col 4	Row 2, Col 5
Row 3, Col 1	Row 3, Col 2	Row 3, Col 3	Row 3, Col 4	Row 3, Col 5

Activate Windows
Go to Settings to activate Windows.

4. Using “table” tag, align the images as follows:

```
<html>

 <head>

<title>p4</title>

</head>

<body>

 <table cellspacing=6 cellpadding=6 align="center">

<tr>

<td><img src="C:\Documents and Settings\All Users\Documents\My
Pictures\Sample Pictures\blue hills.jpg" height="40" width="70">

<td><img src="C:\Documents and Settings\All Users\Documents\My
Pictures\Sample Pictures\sunset.jpg" height="40" width="70">

<td><img src="C:\Documents and Settings\All Users\Documents\My
Pictures\Sample Pictures\blue hills.jpg" height="40" width="70">

<td><img src="C:\Documents and Settings\All Users\Documents\My
Pictures\Sample Pictures\sunset.jpg" height="40" width="70">

 <tr>

<td><img src="C:\Documents and Settings\All Users\Documents\My
Pictures\Sample Pictures\blue hills.jpg" height="40" width="70"> <td
colspan=2><h1>Buy "Tweeties" online

 <td><img src="C:\Documents and Settings\All Users\Documents\My
Pictures\Sample Pictures\sunset.jpg" height="40" width="70">

<tr>

<td><img src="C:\Documents and Settings\All Users\Documents\My
Pictures\Sample Pictures\blue hills.jpg" height="40" width="70"> <td><img
src="C:\Documents and Settings\All Users\Documents\My Pictures\Sample
Pictures\sunset.jpg" height="40" width="70">
```

```
 <td><img src="C:\Documents and Settings\All Users\Documents\My Pictures\Sample Pictures\blue hills.jpg" height="40" width="70">

<td><img src="C:\Documents and Settings\All Users\Documents\My Pictures\Sample Pictures\sunset.jpg" height="40" width="70">

 </table>

 </body>

</html>
```

OUTPUT:



Activate Windows
Go to Settings to activate Windows.

5. Create a menu form using html.

```
<html>
<head>
<title>Menu Order Form</title>
</head>
<body bgcolor="skyblue"text="black">
<h1>
<center>Taj Palace Restaurant<br><br><br>
WELCOME</center><br><br> Restaurant Menu</h1> <h3> <form>
<font color="purple">Main Course Selection:</font><br>
<input type="radio" name="main_course" value="pizza"> Pizza
<br>
<input type="radio" name="main_course" value="pasta"> Pasta
<br>
<input type="radio"  name="main_course" value="salad"> Salad
<br><br><br><br>
<font color="purple"> Add-ons:</font>
<br>
<input type="checkbox"  name="add_ons" value="extra_cheese"> Extra
Cheese
<br>
<input type="checkbox"  name="add_ons" value="garlic_bread"> Garlic
Bread
<br>
<input type="checkbox"  name="add_ons" value="soda"> Soda
```

```
<br><br><br>
<font color="purple">Choose a drink:</font><br>
<select  name="drink">
<option value="water">Water</option>
<option value="juice">Juice</option>
<option value="coffee">Coffee</option>
<option value="tea">Tea</option>
</select>
<br><br><br><br>
<input type="submit" value="Place Order">
</form>
</body>
</html>
```

OUTPUT:



6. Write a program to demonstrate mathematical functions in JavaScript using html.

```
<html>
 <head>
<title> JavaScript Math Object</title>
</head>
<body>
<h2>JavaScript Math Methods</h2>
<p id="demo"></p>
<!-- Script to use math object method -->
<script>
document.getElementById("demo").innerHTML =
 "<p><b>Math.abs(-4.7):</b> " + Math.abs(-4.7) + "</p>" +
"<p><b>Math.min(0, 150, 30, 20, -8, -200):</b> " +
Math.min(0, 150, 30, 20, -8, -200) + "</p>" +
"<p><b>Math.max(0, 150, 30, 20, -8, -200):</b> " +
Math.max(0, 150, 30, 20, -8, -200) + "</p>" +
"<p><b>Math.pow(7, 2):</b>" + Math.pow(7, 2) + "</p>" +
"<p><b>Math.exp(30):</b>" + Math.exp(90) + "</p>" +
"<p><b>Math.sqrt(0.5):</b>" + Math.sqrt(90) + "</p>" +
"<p><b>Math.log(10):</b>" + Math.log(10) + "</p>" +
 "<p><b>Math.ceil(4.4):</b> " + Math.ceil(4.4) + "</p>" +
"<p><b>Math.floor(4.7):</b> " + Math.floor(4.7) + "</p>" +
"<p><b>Math.round(20.7):</b>" + Math.round(20.7) + "</p>" +
 "<p><b>Math.random():</b>" + Math.random() + "</p>" +
```

```
 "<p><b>Math.trunc(2.77):</b>" + Math.trunc(2.77);

</script>

</body>

</html>
```

OUTPUT:

JavaScript Math Methods

Math.abs(-4.7): 4.7

Math.min(0, 150, 30, 20, -8, -200): -200

Math.max(0, 150, 30, 20, -8, -200): 150

Math.pow(7, 2):49

Math.exp(30):1.2204032943178408e+39

Math.sqrt(0.5):9.486832980505138

Math.log(10):2.302585092994046

Math.ceil(4.4): 5

Math.floor(4.7): 4

Math.round(20.7):21

Math.random():0.5734149602396355

Math.trunc(2.77):2

Activate Windows
Go to Settings to activate Windows.

7. Write a program to demonstrate regular expression in JavaScript using html.

```
<html>
<head>
<title>Regular Expression Demo</title>
<style>
body { font-family: Arial, sans-serif; margin: 20px; }
input[type="text"] { padding: 8px; width: 200px; }
button { padding: 10px 15px; background-color: #007bff; color: white;
border: none; cursor: pointer; }
#result { margin-top: 15px; font-weight: bold; }
.valid { color: green; }
.invalid { color: red; }
</style>
</head>
<body>
<h1>Regular Expression Validator</h1>
<p>Enter a username (must start with a letter, contain only letters,
numbers, or underscores, and be 5-15 characters long):</p>
<input type="text" id="usernameInput" placeholder="Enter username">
<button onclick="validateUsername()">Validate</button>
<div id="result"></div>
<script>
function validateUsername() {
const usernameInput = document.getElementById('usernameInput');
 const resultDiv = document.getElementById('result');
```

```
 const username = usernameInput.value;

 const usernameRegex = /^[a-zA-Z][a-zA-Z0-9_]{4,14}$/;

if (usernameRegex.test(username)) {

resultDiv.textContent = `"${username}" is a VALID username.`;
resultDiv.className = 'valid';

 }

else

{

resultDiv.textContent = `"${username}" is an INVALID username.`;
resultDiv.className = 'invalid';

}

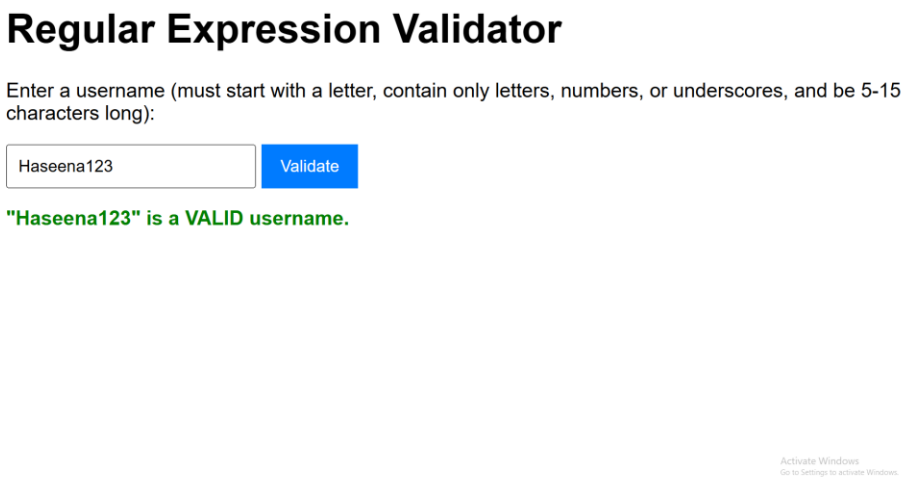
}

</script>

</body>

</html>
```

OUTPUT:



8. Write a program to demonstrate Internal Cascading Style Sheets using html.

```
<html>
<head>
<title>Internal CSS Demonstration</title>
<style>
/* Internal CSS starts here */
body
{
font-family: Arial, sans-serif;
background-color: #f4f4f4;
color: #333;
margin: 20px;
}
h1
{
color: #007bff;
text-align: center;
font-size: 2.5em;
border-bottom: 2px solid #007bff;
padding-bottom: 10px;
}
p
{
font-size: 1.1em;
```

```
line-height: 1.6;
margin-bottom: 15px;
}
.highlight
{
background-color: #ffe0b2;
padding: 5px 10px;
border-radius: 5px;
font-weight: bold;
}
ul
{
list-style-type: square;
margin-left: 20px;
}
li
{
margin-bottom: 5px;
}
</style>
</head>
<body>
<h1>Understanding Internal CSS</h1>
<p>This paragraph demonstrates how <span class="highlight">internal CSS</span> is applied directly within the HTML document's
```

<code><head></code> section using the <code><style></code> tag.

It is useful for styling a single HTML page when the styles are unique to that page and are not intended to be reused across multiple documents.

<h2>Key Features of Internal CSS:</h2>

Defined within the <code><style></code> tags in the <code><head></code> section.

Styles apply only to the HTML elements within that specific document.

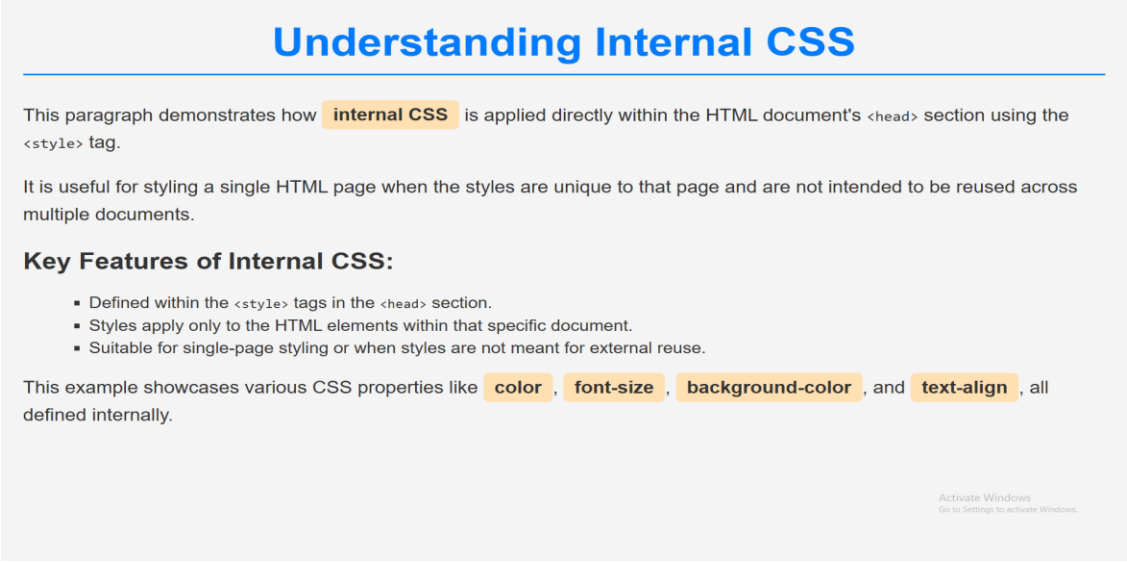
Suitable for single-page styling or when styles are not meant for external reuse.

This example showcases various CSS properties like color, font-size, background-color, and text-align, all defined internally.

</body>

</html>

OUTPUT:



9. Write a program to demonstrate External Cascading Style Sheets using html.

/*save as "styles.css"/

```
/* Body styling */
body
{
font-family: Arial, sans-serif;
background-color: #f4f4f4;
color: #333;
margin: 20px;
}
 /* Heading styling */
h1
{
color: #0056b3;
text-align: center;
}
 /* Class selector styling */
.highlight
{
color: #e67e22;
font-weight: bold;
font-style: italic;
}
 /* ID selector styling */
```

```
#unique-text
{
color: #27ae60;
font-size: 1.2em;
border: 1px solid #27ae60;
padding: 10px;
}
/* Element selector within another element */
div
p
{
background-color: #ecf0f1;
padding: 5px;
border-left: 3px solid #34495e;
}
```

external.html

```
<html>
<head>
<title>External CSS Demo</title>
<!-- Link to the external CSS file -->
<link rel="stylesheet" href="styles.css">
</head>
<body>
<h1>Welcome to the External CSS Demo</h1>
```

```
<p class="highlight">This paragraph is styled using a class from the external stylesheet.</p>

<p id="unique-text">This paragraph has a unique ID and is also styled externally.</p>

<div>

<p>This is a paragraph inside a div, demonstrating element styling.</p>
</div>

</body>

</html>
```

OUTPUT:

